

MODULE 3: BOOKING MANAGEMENT

3.1 Module Name

Booking Management Module

3.2 Description

The Booking Management Module is responsible for managing the ticket booking process within the Smart Event Management and Ticket Booking System. It enables registered users to book tickets for available events in a secure and organized manner. The module retrieves event details, verifies ticket availability, accepts the required booking information from the user, and generates a booking record in the database. It also provides users with the facility to view their booking history, while administrators can monitor and manage all booking records. This module ensures that bookings are processed accurately and efficiently, thereby improving the overall user experience.

3.3 Objectives

The main objectives of the Booking Management Module are:

- To allow registered users to book tickets for available events.
- To store booking information securely in the database.
- To provide users with access to their booking history.
- To enable administrators to manage booking records.
- To maintain accurate and reliable booking information.

3.4 Inputs

The Booking Management Module requires the following inputs:

- User ID
- Event ID
- Ticket Quantity
- Booking Date

3.5 Processing

The module performs the following operations:

1. Verifies that the user is logged into the system.
2. Retrieves the details of the selected event.
3. Checks the availability of tickets.
4. Generates a booking record.
5. Stores the booking information in the database.
6. Redirects the user to the Payment Management Module for payment processing.

3.6 Outputs

The module produces the following outputs:

- Booking Created Successfully
- Booking Confirmation
- Booking History
- Booking Details

3.7 Input Screen

The Booking Form consists of the following fields:

- Event Name
- Event Date
- Ticket Quantity
- Book Ticket Button

3.8 Processing Screen

The system validates the booking information entered by the user, checks ticket availability, creates the booking record, and stores it in the database before proceeding to the payment process.

3.9 Output Screen

After successful booking, the system displays the following information:

- Booking ID
- Event Details
- Number of Tickets
- Booking Status
- Proceed to Payment Button

3.10 Database Collection Used

Collection	Purpose
Event	Retrieves selected event information
User	Retrieves user information
Booking	Stores booking details
Payment	Stores payment information after booking

3.11 Validation Rules

The following validation rules are applied during the booking process:

- The user must be logged into the system.
- The selected event must exist.
- The ticket quantity must be greater than zero.
- The requested number of tickets must not exceed the available event capacity.
- Bookings cannot be created for past events.

3.12 Module Flow

The Booking Management Module follows the sequence below:

User Login → View Events → Select Event → Enter Ticket Quantity → Validate Booking → Save Booking → Proceed to Payment

3.13 Expected Outcome

The Booking Management Module successfully creates and stores booking records in the database after validating the user and ticket availability. It maintains accurate booking information, updates booking records efficiently, and redirects the user to the payment process. The module provides a secure, reliable, and user-friendly booking experience while ensuring data consistency throughout the system.